

Chapter 44 — Prune-Belly Syndrome

Campbell-Walsh-Wein Urology · Part V Paediatric Urology

- Prune-belly syndrome (PBS): a **constellation of anomalies of highly variable severity**.
- **Classic triad**: deficiency of the abdominal wall musculature; bilateral intra-abdominal testes; anomalous urinary tract.
- Urinary tract: variable hydronephrosis, **renal dysplasia**, dilated tortuous ureters, enlarged bladder, dilated prostatic urethra.
- Associated systems: respiratory, gastrointestinal, cardiac, musculoskeletal.
- Spectrum: from **death in the newborn period to minimal involvement**.
- **Principal determinant of long-term survival: severity of the urinary tract anomaly, above all the degree of renal dysplasia**.
- History: **Frölich (1839)** first described the abdominal wall flaccidity → **Parker (1895)** described the full triad → **Osler (1901)** gave the name "prune-belly" from his description of the abdominal wall.
- Synonyms: **triad syndrome, Eagle-Barrett syndrome**, abdominal muscle deficiency syndrome.

Concept — *Survival follows the kidneys, not the belly: the abdominal wall defect that names the syndrome is the least prognostic element of the triad.*

INCIDENCE

- **1 in 29,000 to 1 in 44,000 live births** — similar to bladder exstrophy.
- **95% male**. Females have the abdominal wall deficiency and urinary tract dysmorphism **without any gonadal anomaly**.
- **Heavy disease burden**: admissions and hospital days many-fold greater than in the general paediatric population.
- Higher incidence in **twins, children of African ancestry, and those born to younger mothers**.
- **Declining in developed countries** owing to prenatal diagnosis and termination of pregnancy.
- United States national paediatric inpatient database (newborns 2000–2006): weighted incidence **3.8 per 100,000 live births**.

GENETICS

- **Features suggesting a genetic basis**: high male-to-female ratio, occasional occurrence in male siblings and cousins, increased occurrence in twins.
- Against: **most cases are sporadic with a normal karyotype**.
- Twins: **1 in 23 children with PBS is from a twin pregnancy, yet most twins are discordant** — evidence against a genetic cause. Proposed twin mechanism: **uneven distribution of mesenchyme at primitive streak development, third week**.
- Chromosomal associations: **Turner syndrome, monosomy 16, trisomy 13, trisomy 18**; also **Beckwith-Wiedemann syndrome**. Consensus: a chromosomal abnormality is **the exception, not the rule**.
- Proposed inheritance patterns (none established): X-linked recessive; two-step autosomal dominant mutation; polygenic transmission; **sex-influenced autosomal recessive** in familial PBS.
- **Strongest evidence: multiplex families** (two or more affected). Array copy number variation and next-generation sequencing have identified variants causing PBS or megabladder.

- Gene classes: smooth muscle cell transcription factors; contractile filament proteins or enzymes; cytoskeletal regulatory proteins; neural enzymes or receptors.
- **Key novel genes: myocardin (MYOCD) and filamin A (FLNA).**
- Applications: genetic counselling; **fetal screening of megabladder** for variants. Genotype–phenotype correlation, kindred segregation and pregnancy recommendations still require longitudinal data.

Concept — *The phenotype suggests heredity but most pedigrees do not; the genetic signal lies in smooth muscle genes found in rare multiplex families, consistent with the bladder and ureter being the core lesion.*

EMBRYOLOGY

- **No experimental model** exists, so the mechanism remains elusive; four theories overlap and none is universally accepted.

THEORIES OF EMBRYOGENESIS

Theory	Proposed mechanism
1. Early in utero posterior urethral obstruction	Severe urinary tract dilation, possible fetal ascites and oligohydramnios
2. Primary lateral plate mesoderm defect	Lateral plate mesoderm is precursor of ureters, bladder, prostate, urethra and gubernaculum
3. Intrinsic urinary tract defect	Ureteral dilation leading to fetal ascites
4. Yolk sac defect	Primary defect of the yolk sac

- Pattern: theories divide into **obstructive** (1) and **primary mesenchymal/developmental** (2–4). Trap: the obstruction theory must explain why most patients have **no demonstrable obstruction** postnatally.

Concept — *Every theory must account for a dilated but largely non-obstructed, collagen-rich urinary tract; that observation, not the theories, drives management.*

CLINICAL FEATURES

Genitourinary Anomalies

Kidneys

- Spectrum from **normal parenchyma to dysplasia**. **Dysplasia in about 50%**, variable in degree and laterality.
- Most severe dysplasia: **bladder outlet obstruction without decompression through a patent urachus**.
- Ultrasound signature of dysplasia: **markedly echogenic parenchyma with cortical cysts**.

POTTER TYPE II VERSUS POTTER TYPE IV DYSPLASIA IN PBS

	Potter type II	Potter type IV
Morphology	Few nephrons, parenchymal disorganisation	Cortical and tubular cysts
Implied cause	Renal mesenchymal defect	Outlet obstruction

- Collecting system characteristically dilated, often severely — but **degree of dilation does not correlate with degree of dysplasia**.
- **Calyceal morphology may be well preserved** despite massively dilated ureters and pelvis.
- **Asymmetry**: one kidney severely dysplastic or dilated, the other mildly abnormal.

- Pelviureteric junction (PUJ) obstruction may be primary or secondary, but **non-obstructive hydronephrosis is the rule**.
- **Renal infection, not obstruction, poses the greatest risk to renal function.**

Ureters

- **Dilated, tortuous and redundant. Proximal (upper) ureter less abnormal than distal**, although massive dilation and stenosis can occur at any level.
- Severity of urinary tract abnormality is **not proportional to abdominal wall flaccidity**.
- Histology: **loss of smooth muscle, increased fibrous connective tissue; more normal smooth muscle proximally — critical in ureteral reconstruction.**
- **Raised collagen-to-smooth-muscle ratio, especially in refluxing ureters.**
- Ultrastructure: **fewer thick and thin myofibrils** → poor peristalsis.
- **Vesicoureteric reflux (VUR) in about 75%.**
- Obstruction uncommon but reported at the PUJ and vesicoureteric junction (VUJ); **kinking** at any level may further impair drainage.
- Mechanism of ineffective peristalsis (as in isolated megaureter): **poor wall coaptation** → conduction wave reaches a sparse, weakly contractile muscle population separated by collagen → bolus passes into more dilated, non-propulsive segments → ineffective peristalsis on fluoroscopy → **upper tract stasis** → **infection**.

Concept — The PBS ureter is a **failed pump rather than a blocked pipe**: stasis arises from muscle deficiency, so reconstruction relies on the better-muscled proximal segment and targets stasis and reflux, not obstruction.

Bladder

- **Massively enlarged**, often fixed to the flaccid anterior abdominal wall, with a **urachal pseudodiverticulum**.
- **Urachus patent at birth in 25–30%.**
- Histology: increased collagen-to-muscle ratio in the absence of obstruction; **ganglion cell distribution normal; reduced α 1a-adrenoceptor immunostaining**; fetal bladders show fewer smooth muscle fibres, less type III collagen and fewer nerves.
- **Trigone splayed, ureteric orifices displaced laterally and superiorly** → contributes to the high incidence of VUR.
- **Bladder neck opens widely into a very dilated posterior urethra.**

PRUNE BLADDER VERSUS OBSTRUCTED BLADDER

	Prune bladder (unobstructed)	Obstructed bladder
Wall	Very thick but smooth	Thick and irregular (unlike prune)
Histology	Collagen excess relative to muscle	Smooth muscle hypertrophy (also seen in the obstructed prune bladder)

- Urodynamics: **normal compliance; delayed or absent sensation to void; large capacity.**
- Emptying variable. Residual attributed to **relative outlet obstruction plus inability to generate sufficient detrusor pressure** → "unbalanced voiding".
- **About half void spontaneously** with normal pressures, flow rates and low residuals — but **balanced voiding can deteriorate with growth** → periodic reassessment.

Prostate and Accessory Sex Organs

- **Posterior urethral dilation is due to prostatic hypoplasia**, probably from abnormal mesenchymal-epithelial development.
- Histology: few prostatic elements; fewer epithelial and smooth muscle cells; more connective tissue.
- **Obstructive lesions of the distal posterior urethra in about 20%**: urethral atresia, valves, stenosis, membrane, diverticulum.
- Urethral **angulation during voiding** from lack of prostatic parenchyma.
- Prostatic hypoplasia is a factor in **ejaculatory failure**.
- Vas deferens and seminal vesicles **often atretic**, occasionally dilated or thickened; magnetic resonance imaging (MRI) in adolescents frequently shows **prostatic and seminal vesicle hypoplasia or agenesis**.
- **Epididymis poorly attached to the testis** (as in abdominal undescended testes); possible **discontinuity between efferent ductules and rete testis**.
- **Retrograde ejaculation from an incompetent bladder neck**.

Anterior Urethra

- Usually normal. Commonest anomalies: **urethral atresia or hypoplasia, and megalourethra**.
- **Urethral atresia is often lethal unless associated with a patent urachus**.
- Atresia or hypoplasia postulated to reflect an **unused rather than malformed urethra**.
- Spontaneous bladder rupture with fistula formation reported.
- **Megalourethra is more common in PBS than in any other syndrome**. Proposed cause: **transient in utero obstruction at the junction of glanular and penile urethra**.

SCAPHOID VERSUS FUSIFORM MEGALOURETHRA

	Scaphoid	Fusiform
Deficient tissue	Corpus spongiosum only	Corpora cavernosa and spongiosum
Preserved	Glans and corpora cavernosa	—
On voiding	Ventral urethra dilates	Entire phallus dilates
Mesenchymal deficiency of	Urethral supportive tissues	Urethral folds

- Trap: fusiform is the more severe form — it involves the erectile bodies.

Testes

- **Bilateral intra-abdominal testes over the iliac vessels, adjacent to the dilated ureters** — the most typical finding.
- **Mechanical theory** (distended bladder, intra-abdominal pressure) is doubted because **pseudo-prune patients with the urinary and abdominal anomalies may have descended testes**.
- Fetal **gubernaculum**: altered collagen and elastic fibres — a possible further factor in cryptorchidism.

TESTICULAR HISTOLOGY — ENVIRONMENTAL VERSUS INTRINSIC

Environmental (acquired)	Intrinsic
PBS testes = non-PBS intra-abdominal testes in germ cells, Ad spermatogonia, Leydig cells; germ cells under 1 year similar to age-matched controls → abdominal environment governs later spermatogenic potential. Normal germinal epithelium in fetal and newborn PBS testes	Fewer spermatogonia and Leydig cell hyperplasia in fetal PBS testes; fewer Leydig cells in fetal testes

- **Azoospermia usual in adults; no spontaneous paternity reported**.

- Infertility: **testicular histology + ductal structural defects + prostatic abnormality + retrograde ejaculation.**
- Paternity with normal live births achieved by **sperm retrieval and intracytoplasmic sperm injection (ICSI).** Normal pregnancy with assisted vaginal delivery reported in **affected females.**
- Early childhood orchidopexy followed to adulthood: **normal follicle-stimulating hormone (FSH), luteinising hormone (LH) and testosterone;** some produce semen (antegrade or retrograde) with **spermatozoa in reduced numbers, some motile.**
- **Testicular tumour:** few reported cases; patterns resembling intratubular germ cell neoplasia in infants. Risk probably low (scant germinal epithelium), but **scrotal placement and long-term follow-up** are needed to reduce risk and aid detection.

Concept — *If the abdomen, not the testis, destroys germ cells, then time in the abdomen is the modifiable variable — the basis of early orchidopexy.*

KEY POINTS: CLINICAL FEATURES

- Kidney dysplasia occurs in half of patients.
- Hydroureteronephrosis is often severe, yet calyceal morphology may be well preserved.
- The proximal ureter has more normal muscle than the distal ureter.
- VUR is present in most patients.
- The bladder is large, with a urachal pseudodiverticulum and a wide bladder neck opening into a dilated prostatic urethra.

Extragenitourinary Abnormalities

- **About 75% have non-urinary tract abnormalities.** After the abdominal wall: **cardiac, pulmonary, orthopaedic.**
- **Nearly half are born premature** — a major contributor to comorbidity.

TABLE 44.1 ASSOCIATED NONUROLOGIC ABNORMALITIES IN CHILDREN WITH PRUNE-BELLY SYNDROME (REBUILT)

Frequency (newborn admissions)	Conditions
More than half	Respiratory
Around two in five	Prematurity
About one in four to five	Cardiovascular; gastrointestinal; musculoskeletal; metabolic/endocrine
About one in seven	Sepsis/infectious disease
Uncommon	Other syndrome; head, eyes, ears, nose and throat; immunologic/inflammatory; neurologic; haematologic; dermatologic
Birth weight	Fewer than half weigh over 2500 g; about a quarter under 2000 g

Data from Routh JC, Huang L, Retik AB, Nelson CP. Urology 2010 (newborn males, initial hospitalisation).

- What the table is for: ranking morbidity at birth. Pattern: **respiratory dominates**, and prematurity and low birth weight are common. Trap: figures differ from those in the body text because the denominators differ (below).

FREQUENCY OF ASSOCIATED ANOMALIES — BODY TEXT VERSUS TABLE 44.1

System	Body text	Table 44.1	Reason for difference
Cardiac	10% (structural lesions)	25% cardiovascular	Older series of structural defects versus any cardiovascular diagnosis at newborn admission
Pulmonary	55% of survivors	58% respiratory	Survivors versus all newborn admissions
Gastrointestinal	At least 30%	24%	Different series and case definition
Orthopaedic	Up to 65% (contemporary series)	23% musculoskeletal	Contemporary cohort followed through growth versus newborn admission; many deformities manifest or worsen with growth
Prematurity	Nearly 50%	43%	Rounding of the same order

Abdominal Wall Defect

- The **most characteristic feature** of the newborn. Musculature may be totally absent, but usually **uneven: medial and inferior segments most deficient**.
- At birth: **wrinkled, redundant skin; flank bulging**; viscera visible through the thin wall. Worst areas: skin, subcutaneous fat and a single fibrous layer on the peritoneum.
- **Electromyographic mapping**: inferior and medial segments most consistently affected.
- Electron microscopy: **non-specific** myofilament disarray, Z-line disorganisation, mitochondrial proliferation.
- **Normal spinal anterior horn cells → not neuropathic**.
- Deficiency is patchy and **may be disproportionate to urinary tract severity**.
- With growth: some improve (tonus, fewer wrinkles); others develop a "**potbelly**" with lower rib deformity.
- Gait usually unaffected though may be delayed; children **roll to the side and push up with the arms** to sit.
- **Flaring of the costal margin** from poor lower chest support.
- **Ineffective cough → vulnerability to respiratory illness**.
- **Wound healing good; no tendency to infection or incisional hernia**.

Cardiac Anomalies

- **About 10%**: patent ductus arteriosus (PDA), atrial and ventricular septal defects (ASD, VSD), tetralogy of Fallot.
- **Cardiac problems at birth may take precedence over urological issues**.

Pulmonary Abnormalities

- Occur at any age.
- **Pulmonary hypoplasia from severe oligohydramnios** (renal dysplasia or severe outlet obstruction) → may cause newborn death.
- **Pneumothorax and pneumomediastinum**, with or without hypoplasia.
- Significant pulmonary difficulty in over half of survivors; **nearly half of newborns need intubation and ventilation**.
- Inability to raise intra-abdominal pressure → **recurrent pneumonia and lobar atelectasis**.
- Acute illness or **anaesthesia readily precipitates respiratory infection and insufficiency**, often on a background of chronic bronchitis.
- **Restrictive lung disease** from scoliosis and rib cage deformity, **commonest pectus excavatum**.

Gastrointestinal Abnormalities

- **At least 30%.**
- Majority from **incomplete midgut rotation** → **wide mesentery** → **excess bowel mobility** → malrotation, volvulus, atresia, stenosis.
- **Splenic torsion** from abnormal mesenteric fixation.
- Also: omphalocele, gastroschisis, situs inversus abdominis, VACTERL association (vertebral defects, anal atresia, cardiac defects, tracheo-oesophageal fistula, renal anomalies, limb abnormalities), **anorectal malformation including imperforate anus**.
- Poor intra-abdominal pressure → **lifelong constipation, acquired megacolon**.

Orthopedic Abnormalities

- **Up to 65%** in contemporary series; **second in frequency only to the genitourinary tract and abdominal wall**.

CAUSE OF MUSCULOSKELETAL DEFECTS — TWO VIEWS

Mesenchymal	Compressive
Abnormal mesenchymal development at 6 weeks' gestation	Oligohydramnios compression — favoured because many deformities are unilateral

- Oligohydramnios-related: **talipes equinovarus (commonest)**, hip dysplasia, congenital scoliosis.
- **Dimpling of the lateral knees**: oligohydramnios sign, frequent in PBS.
- Distended bladder compressing the **external iliac vessels** → lower limb hypoplasia, absence or amputation.
- **Manifest or worsen with growth**: renal osteodystrophy, hip dislocation, scoliosis, pectus excavatum or carinatum.

Oral Abnormalities

- Dental and bone abnormalities reported.

Concept — Two mechanisms generate most extragenitourinary morbidity: **oligohydramnios** (pulmonary hypoplasia, limb deformity, knee dimples) and **absent abdominal wall pressure** (ineffective cough, atelectasis, constipation).

KEY POINTS: ASSOCIATED ABNORMALITIES

- Megalourethra is more common in PBS than in any other syndrome; in the scaphoid variety the ventral urethra dilates on voiding, in the fusiform the entire phallus.
- Infertility reflects testicular histology, ductal defects, and prostatic and vesicular abnormalities with retrograde ejaculation.
- Pulmonary hypoplasia from severe oligohydramnios (renal dysplasia or severe outlet obstruction) may cause newborn death.
- Abdominal flaccidity has no direct relationship with urological severity and does not impair short- or long-term survival; with orthopaedic abnormalities, if significant and untreated, it may reduce quality of life in adolescents and adults.

PRESENTATION

Prenatal Diagnosis

- Fetal hydronephrosis is diagnosed accurately in the **second trimester** and is present in **about 1% of pregnancies**; its **cause cannot always be determined** (accuracy estimated 30–85%).
- **PBS mimics other causes of bladder outlet obstruction — posterior urethral valves, megacystis–megaureter syndrome.**
- Diagnosed as early as **11–14 weeks**, yet the classic findings — **hydroureteronephrosis, distended bladder, irregular abdominal circumference** — are **not consistently seen at 30 weeks**.
- **Early fetal ascites** has been suggested to correlate with PBS.
- Most have **no demonstrable obstruction**; **degree of prenatal hydroureteronephrosis does not correlate with postnatal renal function.**
- Sonographic appearances: massively dilated bladder filling the abdomen with **absent amniotic fluid**; bladder reaching the kidneys; urachal diverticulum; **elongated, dilated posterior urethra**; echogenic hydronephrotic kidneys. A **dilated prostatic urethra indicates outlet obstruction from urethral atresia.**

Neonatal Presentation

- **Flaccid, wrinkled abdominal wall at birth is diagnostic on sight**, whether or not suspected prenatally.
- Reinforcing signs: **bilateral impalpable testes, knee dimples.**
- **Pulmonary and cardiac problems take precedence**: without true outlet obstruction (as in urethral atresia), hydroureteronephrosis is **not immediately life-threatening.**

Spectrum of Disease

Woodard's Classification

TABLE 44.2 SPECTRUM OF PRUNE-BELLY SYNDROME (REBUILT)

Category	Kidneys	Lungs	Defining features
I	Renal dysplasia	Pulmonary hypoplasia	Oligohydramnios, Potter features, urethral atresia
II	Minimal or unilateral dysplasia; may progress to renal failure	No hypoplasia	Full triad
III	No dysplasia; stable function	No hypoplasia	Incomplete or mild triad; mild-to-moderate uropathy

Modified from Woodard JR. Prune belly syndrome. In: Kelalis PP, King LR, Belman AB, eds. Clinical Pediatric Urology. Philadelphia: WB Saunders; 1985.

- What the table is for: **triage by lung and kidney**. Pattern: moving down, pulmonary hypoplasia disappears first, then dysplasia. Trap: the abdominal wall is not a sorting criterion.
- **Category I**: marked oligohydramnios from bilateral dysplasia with severe ureteral or outlet obstruction → pulmonary hypoplasia and skeletal anomalies. **Most urethral atresia falls here** — except atresia with a patent urachus. Non-stillborn infants usually die within days (pulmonary hypoplasia) or later (renal failure). **About 20% of newborns with PBS die perinatally.**
- **Category II**: full triad, moderate or unilateral renal insufficiency, moderate-to-severe hydroureteronephrosis; pulmonary hypoplasia not prominent. Lungs normalise; renal function stabilises at or below normal — but **early azotaemia if significant stasis or obstruction**; with hydroureteronephrosis,

VUR, residual urine and infection, further impairment. **The category in which management controversy lies.**

- **Category III:** mild or incomplete forms; **the majority;** some hydroureteronephrosis with **stable renal function;** no pulmonary insufficiency.
- Poor correlation between abdominal wall deficiency, hydronephrosis and dysplasia: some have markedly dilated tracts with little dysplasia and retain normal function → **abdominal appearance or degree of hydronephrosis has little bearing on long-term prognosis.**

WOODARD CATEGORIES — COURSE AND MANAGEMENT

Category	Course	Urological management
I	Perinatal death from pulmonary hypoplasia or renal failure	Supportive care only; simple bladder (catheter) drainage — intervention does not alter outcome
II	Variable; risk of progressive renal insufficiency	Individualised; functional imaging; conservative versus reconstructive debate
III	Balanced hydronephrosis, good renal function	Surveillance; surgery only for recurrent urinary tract infection (UTI) or upper tract deterioration; orchidopexy in first year

Incomplete Syndrome

- Males lacking the full triad but sharing other features.
- Delayed diagnosis (absent obvious abdominal signs) → recurrent UTI → **late renal failure in a high proportion;** hence close observation, monitoring and selective intervention.
- Those without abdominal wall signs may present **as late as adulthood with renal failure and hypertension;** most adult presenters eventually develop UTI from chronic stasis.

CLASSIC PBS VERSUS PSEUDO-PRUNE

	Classic	Pseudo-prune (incomplete)
Abdominal wall	Flaccid, wrinkled	Very slight or unilateral flaccidity; no wrinkling
Testes	Bilateral intra-abdominal	Unilateral undescended or partially descended
Uropathy	Variable	Variable
Diagnosis	At birth	Delayed, sometimes to adulthood
Renal risk	By category	Relatively high late renal failure

Concept — *The obvious phenotype protects the kidneys by forcing early surveillance; the subtle phenotype harms them through delayed diagnosis and untreated infection.*

Female Syndrome

- **5% of PBS;** most have abdominal wall deficiency and an abnormal urinary tract; some have the wall defect with a **normal urinary tract. No gonadal anomaly.**
- In the largest series: **outlet obstruction common, anorectal anomalies frequent,** and **high newborn mortality, similar to males.**

RUBACE Classification

- Phenotypic **severity score for living patients in Woodard categories II and III.**
- **RUBAC** = Renal, Ureter, Bladder, Abdominal wall, Cryptorchidism (the triad; **maximum 16**). **E** = Extragenitourinary, scored by organ system or by an existing congenital syndrome or association ("PBS-plus") (**maximum 15**).
- **RUBAC + E = RUBACE;** stated range **5–31. Normal anatomy scores zero** in any subcategory.

- Goals: group similar cases for **genetic identification**; genotype–phenotype correlation; **monitor deterioration with age**; match severity groups for surgical and medical outcome analysis.

FIGURE 44.18 THE PBS RUBACE SEVERITY SCORING RUBRIC (REBUILT)

Domain (max)	1 point	2 points	3 points	4 points
Renal (6)	Chronic kidney disease (CKD) G1–G2: structural damage (dilation, dysplasia, scarring), glomerular filtration rate (GFR) ≥ 60	G3a–b: GFR 30–59	G4: GFR 15–29	GFR <15, dialysis, or post-transplant
Young-age points: +2 if under 2 years; +1 if under 13 years (brings renal maximum to 6)				
Ureteral (3)	Low-grade or absent VUR, or distal ureter 4–10 mm	Persistent high-grade VUR (unoperated), post lower ureteral reconstruction, or distal ureter 1–2 cm	High-grade VUR despite surgery , distal ureter >2 cm, or current diverting ureterostomy/nephrostomy	—
Bladder/outlet (3)	Urotherapy needed to empty, or bladder 100–200% of normal	Catheterisation or drugs needed to empty, bladder >200% of normal, or antibiotic prophylaxis for recurrent UTI	Current surgical diversion (tube, vesicostomy, appendicovesicostomy), urethral atresia, or megalourethra needing repair	—
Abdominal wall (2)	Can do a sit-up (before or after abdominoplasty); minimal laxity; mild prune appearance	Severe laxity, classic prune appearance, or post-abdominoplasty and cannot sit up	—	—
Cryptorchidism (2)	Unilateral or palpable undescended testis (current or past), or female	Bilateral non-palpable testes (current or past)	—	—
Neurologic (3)	—	—	Seizures, tethered cord, spina bifida, hearing loss, intellectual delay, autism, severe condition not otherwise specified	—
Cardiac (3)	Septal defects (PDA, ASD, VSD) that resolved spontaneously	Small unresolved septal defects, minimal left-to-right shunt, or PDA needing surgery	Severe or cyanotic disease, tetralogy of Fallot, left-sided obstructive lesions, reversed shunt, congestive heart failure	—
Gastrointestinal (3)	Mild–moderate constipation on diet or laxatives	Malone antegrade continence enema (MACE) or caecostomy, or malrotation after Ladd procedure	Surgical bowel diversion (imperforate anus, anal atresia), gastrointestinal malignancy (e.g. hepatoblastoma), severe condition	—
Musculoskeletal (3)	Mild facies, abnormal digits without loss of function, rib flaring	Scoliosis, hip dysplasia, club or rocker-bottom foot, genu valgum, pectus excavatum or carinatum	Arthrogryposis, muscular dystrophy, severe malformation	—

Domain (max)	1 point	2 points	3 points	4 points
Respiratory (3)	Minimal support at birth, some cough difficulty, mild intermittent asthma or reactive airway disease	Persistent/moderate asthma, ≥ 3 respiratory infections per year, or ≥ 2 admissions for pneumonia	Ventilator-dependent >1 week, tracheostomy, pneumothorax, severe asthma	—

Modified from Wong DG, Arevalo MK, Passoni NM, et al. Phenotypic severity scoring system and categorisation for prune belly syndrome: application to a pilot cohort of 50 living patients. *BJU Int* 2019;123(1):130–139.

- What the rubric is for: **quantifying morbidity in survivors**. Pattern: the triad (RUBAC) carries slightly more weight than extragenitourinary disease; renal is the heaviest single domain. Traps: **female patients score 1 for cryptorchidism**; the abdominal wall maximum is only 2.

Concept — Woodard sorts newborns by survival (lung and kidney); RUBACE grades living children by cumulative morbidity so that category II and III outcomes can finally be compared.

EVALUATION AND MANAGEMENT

Prenatal Management

- Some advocate **in utero intervention** for dilation and oligohydramnios; others **termination**.
- **Termination is difficult to justify**: the cause of prenatal hydronephrosis cannot be diagnosed precisely and postnatal renal function cannot be predicted from dilation — **except in rare early, severe oligohydramnios**.
- **Prenatal intervention: no proven benefit for postnatal renal function**.
- **Justified only in: urethral atresia with progressive oligohydramnios; decompression to prevent dystocia**.

Neonatal Management

- **Team**: neonatologist, nephrologist, urologist; cardiologist as indicated; early orthopaedic review for significant limb anomalies.
- **First priority: cardiac and respiratory**. **Immediate chest radiograph** for pneumothorax, pneumomediastinum and pulmonary hypoplasia.
- **Early urological intervention only for bladder outlet obstruction**: bladder catheter or, eventually, **percutaneous suprapubic tube** in the neonatal intensive care unit.
- Initial **creatinine reflects maternal function** → the **trend over days to weeks** better predicts long-term function.
- **Serum creatinine, blood urea nitrogen (BUN), electrolytes**: detect acidosis and electrolyte disturbance.
- **Baseline creatinine <0.7 mg/dL predicts adequate renal function through childhood**, absent repeated pyelonephritis.
- **Renal and bladder ultrasonography once stable**: parenchymal thickness, echogenicity, cortical cysts, degree of dilation.
- **Avoid UTI** (stasis plus compromised function): **circumcision** if no structural penile abnormality; **antibiotic prophylaxis**, especially before instrumentation including the first voiding cystourethrogram (VCUG).
- VCUG: avoid purposeless instrumentation, but **indicated with renal insufficiency** to distinguish **outlet obstruction from upper tract stagnation** and to assess emptying. **VUR found in up to 70%**.
- **VCUG not needed early** if renal function is normal and bladder drainage via urethra or **patent urachus** is adequate.
- **Strict sterile technique** for any instrumentation — a static system is easily inoculated.

KEY POINTS: NEONATAL MANAGEMENT

- A team of neonatology, nephrology, urology and others such as cardiology is necessary.
- A chest radiograph for pneumothorax, pneumomediastinum and pulmonary hypoplasia is necessary.
- Baseline renal assessment: renal and bladder ultrasonography, BUN, creatinine and electrolytes.
- VCUG in the neonatal period only for renal insufficiency or evidence of outlet obstruction, and only after antibiotic prophylaxis.
- Circumcision is advisable in the absence of a structural penile abnormality.
- Early intervention for outlet obstruction, preferably with a percutaneous suprapubic tube.

Infant Management

- Neonates are managed by Woodard category (Table 44.2).
- **Category I: supportive care only**; urinary tract intervention no more than simple bladder drainage — fate usually cannot be altered.
- **Category III**: rarely early urinary tract intervention (balanced hydronephrosis, good function). Regular monitoring of **dilation (ultrasonography)**, **renal function (serum creatinine)** and **UTI**. **Cryptorchidism corrected in the first year**. Persistent VUR: prophylaxis, monitoring, **surgery if symptomatic**. **Abdominoplasty** if flaccidity persists or worsens.
- **Category II: individualised** (variable severity of each facet); much controversy. Renal function and drainage assessment required in those with renal insufficiency.

IMAGING AND FUNCTIONAL STUDIES IN PBS

Study	Role	Caveat
Renal and bladder ultrasonography	First-line baseline and surveillance	Anatomy only
VCUG	Outlet, emptying, VUR	Only with renal insufficiency or suspected outlet obstruction ; under antibiotic cover
^{99m} Tc-dimercaptosuccinic acid (DMSA)	Best test of parenchymal (differential) function	Perform at 4–6 weeks to avoid transitional neonatal physiology
^{99m} Tc-mercaptoacetyltriglycine (MAG3) or diethylenetriaminepentaacetic acid (DTPA)	Outflow obstruction with comparative function	Limited by poor function and massive dilation
Magnetic resonance urography (MRU)	Calyceal diverticula, abnormal pelvic anatomy, characterisation of dysplasia ; more detail than ultrasound, urography or nuclear scans	Gadolinium (rare nephrogenic systemic fibrosis), cost, availability, sedation
Excretory urography	Dramatic anatomical images	No comparative function; radiation — no longer used
Whitaker antegrade perfusion	Selective assessment of obstruction	Rarely performed now

- Pattern: **DMSA for function, MAG3/DTPA for drainage**. Trap: a DMSA before 4–6 weeks misleads; nuclear drainage curves are unreliable in massive dilation with poor function.

Clinical Treatment

- Conservative school: **close surveillance, medical treatment of bacteriuria, surgery only for proven obstruction or intractable infection**.

- VUR: no reason to consider it less important in PBS — **correction of high-grade reflux is prudent.**
- Conservatively managed series followed from infancy: most remained well apart from occasional UTI, with normal voiding and renal function. **Category III patients are candidates for this approach.**
- **Spontaneous improvement** of ureteral appearance and function may occur with growth and ureteral elongation; some with gross abnormality survive decades untreated.
- Yet **progressive uropathy** is well recognised; many ultimately need **renal transplantation.**

Surgical Treatment

- Origin: historical compilation (1950–1970) showed that **most category II infants died**, with or without surgery → a more aggressive approach was adopted.
- Rationale: **infection and progressive renal insufficiency are the chief threats** → reconstruct to normalise anatomy and function.
- Components: ureteral shortening, **tailoring or infolding**, ureteric reimplantation, reduction cystoplasty; rarely, **removal of symptomatic poorly functioning dysplastic or hydronephrotic kidneys.**
- **Delay reconstruction until at least 3 months of age for pulmonary maturation.**
- Outcome: stable imaging, stable creatinine, fewer infections.
- **Comprehensive single-stage surgery** — urinary tract reconstruction with orchidopexy, abdominoplasty and circumcision — **does not increase morbidity**; in long follow-up most maintain normal creatinine, a minority develop insufficiency or need transplantation.
- Comparisons are hampered by scant long-term data, inconsistent severity assignment and variable natural history; **individualisation of care** is emphasised; controversy over category II will persist until stratification such as RUBACE allows accurate selection.
- Surgery falls into three groups: **urinary tract, orchidopexy, abdominal wall.**

CONSERVATIVE VERSUS RECONSTRUCTIVE PHILOSOPHY

	Conservative (limited intervention)	Reconstructive (early comprehensive)
Premise	Dilation is non-obstructive and may improve with growth	Stasis, reflux and infection drive renal loss
Operate when	Proven obstruction or intractable infection	Progressive dilation, high-grade VUR, recurrent infection, declining function
Best suited	Category III	Selected category II
Weakness	Progressive uropathy in some	Heterogeneous, retrospective evidence

Urinary Tract

- **Indications for reconstruction:** progressive or severe hydroureteronephrosis; high-grade VUR; recurrent upper tract infection; true obstructive uropathy; progressive renal failure.
- **Temporary diversion** for the very young or very ill child.

Urinary Diversion

- **Supravesical Urinary Diversion:** for early repeated upper tract infection or deteriorating function. **Vesicostomy usually suffices**; rarely, PUJ, mid-ureteric or VUJ obstruction requires **cutaneous pyelostomy or distal ureterostomy. Avoid proximal ureterostomy — it sacrifices the normal proximal ureter needed for later reconstruction.**
- **Cutaneous Vesicostomy:** indications — **acute renal failure, urosepsis, outlet obstruction from urethral atresia with limited urachal patency. Procedure of choice for temporary diversion: Blocksom technique.** Excise a large urachal diverticulum at the same time. **Create a larger-than-normal stoma — stenosis is**

common (low intra-abdominal pressure). A double-J stent may be passed through the urethra alongside for progressive urethral dilation. Decompression may unmask cortical cysts on ultrasound.

Upper Urinary Tract Reconstruction

- **Ureteroplasty and ureteroneocystostomy:** remodelling remains **controversial**. Best for **repeated non-suppressible upper tract infection** or **progressive anatomical or functional deterioration** from dilation and tortuosity, often with VUR. **Goal: reduce stasis and correct massive reflux.**
- Keys: meticulous technique; **preserve the upper few centimetres of proximal ureter** (less dilated); tailoring or infolding is often still needed; **submucosal tunnelling in the abnormal bladder is difficult.**
- **Pyeloplasty:** secondary PUJ obstruction → **proximal ureterolysis** preserving ureteral vascularity may decompress the pelvis; true mechanical obstruction → **non-dismembered pyeloureteral anastomosis** between dilated pelvis and a normal upper ureteral segment.
- Timing: excellent results reported neonatally and in older children, but **extensive reconstruction is no longer recommended before 3–6 months**. Comprehensive surgery series spanning decades report **functional stabilisation of the urinary tract in up to 90%.**

Lower Urinary Tract Reconstruction

- **Reduction Cystoplasty:** poor contractility → incomplete, infrequent emptying → stasis and VUR → reduce and remodel into a **spherical** shape to direct contraction. Techniques range from **urachal diverticulum excision** to mucosal excision with **overlapping muscle flaps**. **Capacity and residuals recur** → justified only to excise a large urachal diverticulum or as part of wider reconstruction. **Clean intermittent catheterisation (CIC)** per urethra or via appendicovesicostomy gives better long-term emptying until voiding pressure improves with age or abdominoplasty.
- **Internal Urethrotomy:** normal sphincteric resistance implicated in "unbalanced" voiding; urethrotomy improved flow, residual and upper tract appearance. Considered for high residuals, worsening hydroureteronephrosis or VUR with recurrent infection, but **no sustained long-term success**. Techniques: **Otis urethrotome to 24–30 Fr**, one or two anterior/anterolateral incisions; or **direct-vision urethrotomy at the distal prostatic urethra**. **No incontinence resulted**. No further experience published; **not a procedure for infants today**.
- **Anterior Urethral Dilation or Reconstruction:** atresia or hypoplasia; some survive without intervention but often need treatment to improve emptying. **Progressive gentle dilation**, in situ or **through-and-through via a vesicostomy**; not uniformly successful. **Hydrodistension** (one to three sessions) improves calibre in hypoplasia. **No formal graft urethroplasty reported in PBS.**
- **Megalourethra repair** (fusiform or scaphoid): circumferential subcoronal incision → **penile degloving** → expose megalourethra → **tailor by excision or infold** → reconstruct over an appropriately sized catheter → excise redundant prepuce → approximate skin.
- **Circumcision:** as in other boys at UTI risk (posterior urethral valves, VUR), **reduces recurrent UTI**; incorporated in all reconstructions, orchidopexies and abdominoplasties in one large series.

LOWER URINARY TRACT PROCEDURES — STANDING

Procedure	Rationale	Verdict
Reduction cystoplasty	Smaller, spherical bladder empties better	Capacity recurs ; only for large diverticulum or with wider reconstruction
Internal urethrotomy	Lower outlet resistance in unbalanced voiding	Short-term gains, no sustained success , abandoned
CIC (urethral or Mitrofanoff)	Mechanical emptying	Better long-term emptying

Procedure	Rationale	Verdict
Urethral dilation / hydrodistension	Open atretic or hypoplastic urethra	Variable success; no graft urethroplasty reported
Megalourethra repair	Tailor redundant urethra	Degloving and tailoring/infolding
Circumcision	UTI reduction	Routine with reconstruction

Orchidopexy

- Timing set by the need for **early treatment of the undescended testis** and by each child's need for temporary or definitive reconstruction.
- Justification for early surgery: **germ cells present in infancy**; motile spermatozoa found in semen and post-ejaculatory urine of adults after infant orchidopexy; **excellent prognosis for pubertal hormonal function**; **testicular malignancy risk**.
- Testes lie intra-abdominally on a **broad mesorchium over the iliac vessels or dilated ureters** → **standard inguinal approaches usually fail**.
- **Transabdominal Orchidopexy**: neonatally and **up to 6 months**, transabdominal mobilisation of spermatic vessels usually allows scrotal placement. **Bilateral transabdominal orchidopexy at about 6 months is the approach of choice**, often combined with vesicostomy, urinary reconstruction or abdominoplasty. Vascular pedicle may or may not be preserved.
- When not achievable in the first months: **(1) Fowler-Stephens orchidopexy**; **(2) staged Fowler-Stephens orchidopexy**; **(3) microvascular autotransplantation**. Meta-analysis: **staged Fowler-Stephens more successful than single-stage**. After first-stage Fowler-Stephens, **vasal collateral vascularity increases** by the second stage.
- **Laparoscopic Orchidopexy**: when no other abdominal surgery is required — primary (vessels preserved) or Fowler-Stephens in one or two stages; satisfactory long-term outcomes. Technical adjustment: **port placement — little resistance to cannula insertion** in the lax wall.

ORCHIDOPEXY OPTIONS IN PBS

Approach	When	Key feature
Transabdominal (open), bilateral	Neonate to ~6 months; choice at ~6 months	Vessels mobilised; combined with other abdominal surgery
Fowler-Stephens, single-stage	After early window	Vessel division; relies on vasal collaterals
Staged Fowler-Stephens	After early window	Collaterals develop between stages; higher success
Microvascular autotransplantation	After early window	Microvascular anastomosis
Laparoscopic (primary or Fowler-Stephens)	No other abdominal surgery needed	Port placement adjusted for lax wall

Concept — *Early in infancy the spermatic vessels still mobilise enough for a primary transabdominal repair; beyond that window the testis must depend on vasal collaterals, which is why staging outperforms single-stage division.*

Reconstruction of the Abdominal Wall

- Mild deficiency may **improve with maturation**; moderate-to-severe laxity leaves a **physically and psychologically crippling defect**.
- **Elasticised corset**: improves clothed appearance and respiratory and gastrointestinal function, even neonatally; inconvenient long-term, especially in **warm climates**.

- **Cosmetic benefit agreed; functional benefit (bladder, bowel, lung) controversial** — improved emptying after reconstruction was confounded by concomitant urinary remodelling. Potential gains: more effective cough, better defecation.
- **Timing dictated by other planned surgery**, particularly upper tract remodelling; otherwise any time — as early as **6 months with transabdominal orchidopexy**. In infancy, **postoperative ventilation** may be needed.

ABDOMINOPLASTY TECHNIQUES COMPARED

Technique	Incision	Umbilicus	Fascial handling	Peritoneum	Note
Randolph	Transverse: 12th rib → pubic symphysis → opposite 12th rib	—	Full-thickness excision of skin, lower musculature and peritoneum; healthy fascia to anterior iliac spines, pubic tubercle, inferior fascia	Excised	Based on EMG mapping (infraumbilical worst); waistline achieved but lateral bulging persists
Ehrlich	Vertical midline	Preserved on inferior epigastric pedicle	Skin raised off muscle; vest-over-pants advancement of each side to contralateral flank; lateral muscles preserved	—	Excellent long-term results
Monfort	Vertical elliptical, xiphoid to pubis	Circumscribed, left in situ	Dissect to anterior axillary line; vertical fascial incisions lateral to superior epigastric arteries → central fascial bridge ; lateral fascia advanced over it	Opened via lateral incisions	Excellent exposure for concomitant urinary surgery and orchidopexy ; thickens wall
Dénes	Fusiform longitudinal skin/subcutaneous resection	Kept in the wide flap; buttonhole	One eccentric elliptical xiphopubic fascial incision on the more flaccid side → wide and narrow flaps; wide flap sutured to inner side of narrow; narrow flap overlapped onto wide (double-breasted)	Opened	Modification of Ehrlich; appearance and tonus improved in the great majority , all after reoperation
Furness	Midline, limited dissection	—	Double-breasted plication of lateral fascia in the midline ; no lateral dissection or central plate incision	Not entered	For when no intra-abdominal surgery is planned
Fernández-Bautista	Midline	Neoumbilicoplasty with island flap rotated on itself	Complete fascial overlap plication; mesh if fascia and muscle absent or insufficient	Not entered	Modification of Furness
Muscle transfer	—	—	Rectus femoris ± sartorius transfer, or Ehrlich plus rectus femoris transposition	—	Subjective gains in balance, ambulation, cough; donor-site cost

Technique	Incision	Umbilicus	Fascial handling	Peritoneum	Note
Laparoscopic / robotic	Elliptical skin excision (laparoscopic-assisted)	—	Pneumoperitoneum identifies weak areas → plication (Furness or Monfort-based). Robotic adult repair: preperitoneal flap, three parallel mattress sutures, synthetic mesh covered by peritoneum	Protected / preperitoneal	Robotic approach for adult lateral bulging after childhood Randolph

- Monfort abdominoplasty with concomitant reconstruction (sequence): tent the wall to delineate redundancy → outline skin incisions with separate circumumbilical incision → excise skin (epidermis and dermis) with diathermy → incise the central plate at the lateral rectus border from superior to inferior epigastric vessels → transperitoneal genitourinary surgery: **preserve only the more normal proximal ureter**, excise the urachal diverticulum, **transtrigonal reimplantation ± tapering**, two-layer bladder closure with ureteric stents and cystostomy → score parietal peritoneum over lateral musculature → suture central plate edges to the scored line → bring lateral flaps together in the midline over closed suction drains → close skin around the umbilicus.
- Dénes closure (sequence): estimate extent → mark limits → elliptical excision of skin and subcutaneous tissue preserving umbilicus → eccentric elliptical fascial incision on flaccid side → suture broad flap to inner side of narrow flap → fix narrow flap over broad flap with umbilical buttonhole → approximate subcutaneous tissue and skin.
- Trap: **Furness keeps the peritoneum closed; Monfort opens it** — choose by whether intra-abdominal surgery is planned.

Concept — Abdominoplasty techniques differ on two axes — **transverse versus vertical** and **transperitoneal access versus extraperitoneal plication** — and the second is decided by whether urinary reconstruction or orchidopexy is being done at the same sitting.

KEY POINTS: SURGICAL RECONSTRUCTION

- Upper urinary tract reconstruction is controversial but clearly indicated for declining renal function with recurrent upper tract infection or progressive hydronephrosis.
- Orchidopexy is best performed early, giving the best prospect of a successful single-stage procedure and of germ cell preservation.
- Abdominal wall reconstruction, besides cosmetic and psychosocial benefit, has shown improved bladder emptying, more effective cough and improved defecation.
- Comprehensive surgery including all of the above, with circumcision, is feasible in most patients.

LONG-TERM OUTLOOK

Quality of Life

- Survival and quality of life **much improved** through medical, surgical and urodynamic advances; **perinatal mortality remains high in category I** (prematurity, dysplasia, pulmonary hypoplasia).
- Health-related quality of life **reduced across physical, emotional, social and school functioning**; **caregivers** also report lower quality of life.
- Orthopaedic, gastrointestinal and cardiopulmonary conditions worsen quality of life; **treating non-genitourinary conditions improves it**.

- The **paediatric urologist, as principal physician**, must recognise and manage non-genitourinary comorbidity through puberty into adulthood.

Renal Function

- **Nadir creatinine <0.7 mg/dL at 1 year → stable renal function through childhood**, unless further insult from pyelonephritis or obstruction.
- Reconstruction maintains stable function in most, but **periodic imaging, urine cultures and prompt UTI treatment** are essential — infection risk is constant (dilation, stasis, CIC).
- **Up to 30% develop chronic renal failure** in childhood or adolescence, generally those with impaired function at first evaluation.
- **Dysplasia and polyuria** also harm long-term function.
- **Transplantation** ensures normal growth and development; **results equal age-matched groups**; recipients can **retain their native urinary tracts**.
- **Individualisation**: some need major reconstruction, others almost none; **lifelong surveillance** because dynamics change.

Bladder Function

- **Normal compliance, decreased sensation and contractility** → large retained volumes, progressive capacity, significant residual.
- Aggravated by **lack of abdominal support, polyuria and poor adherence to timed voiding**.
- **All patients**, even after reconstruction and abdominoplasty, are trained early: **timed voiding, double or triple voiding, Valsalva or Credé manoeuvres**.
- Progressive unbalanced voiding with failed training → **intermittent or nocturnal catheterisation is mandatory** (urethral or Mitrofanoff).
- Effective voiding eventually follows growth, improved proprioception and abdominal support; a minority after reconstruction needed CIC, and **some regain normal voiding as they grow**.

Pulmonary and Gastrointestinal Function

- Infant and early-childhood pulmonary problems **decline with age** as thoracic and abdominal tone improves, but **asthma and reactive airway disease are more prevalent** in older patients.
- **Constipation in more than half**; constant management required.

Growth and Musculoskeletal Development

- **Normal growth expected with preserved renal function**, although growth retardation without renal compromise occurred in a third in one series.
- Musculoskeletal problems may improve or persist into adulthood.
- Abdominal flaccidity may improve with age, but limitations persist; **corsets; swimming** strengthens musculature without strain; adults with **lateral bulging: robotic synthetic mesh**.
- **Pectus excavatum rarely compromises lung function** in older children; thoracoplasty for cosmesis in early adulthood.
- **Significant scoliosis and lordosis: early orthopaedic management**.
- Lower limb anomalies detected in infancy, some best corrected then; hip and ankle deformities may appear later and need surgery.
- Families report that **correcting musculoskeletal problems (abdominoplasty or orthopaedic surgery) is the commonest way care improved quality of life**.

Sexual Function and Fertility

- **Normal secondary sexual development** — testicular hormone production preserved.
- **Normal libido and erections** when penile anatomy is preserved.
- **Primary infertility assumed** (poor sperm quality, retrograde ejaculation); **no spontaneous paternity reported**.
- **Early orchidopexy preserves motile spermatozoa** in limited numbers → **assisted reproduction feasible**.

Concept — *Long-term care is a contest against stasis: whatever reduces residual urine and infection (timed voiding, catheterisation, reconstruction) protects the kidneys, while dysplasia and polyuria set the ceiling surgery cannot raise.*

KEY POINTS: LONG-TERM FOLLOW-UP

- Up to one-third may develop chronic renal insufficiency during childhood or adolescence.
- Recurrent pyelonephritis, renal dysplasia and polyuria are the major factors in renal decline.
- Adequate bladder emptying is necessary to avoid recurrent infection and prevent upper tract deterioration associated with polyuria.
- Pulmonary and gastrointestinal abnormalities require constant care in a significant proportion of older patients.
- Abdominal tone improves spontaneously with growth in most, but skeletal abnormalities often need surgical correction.
- Normal sexual development is expected after orchidopexy, but with reduced sperm quality and retrograde ejaculation no spontaneous paternity has been reported.